

Benchmark Specifications

1 Duffing's oscillator

1.1 System

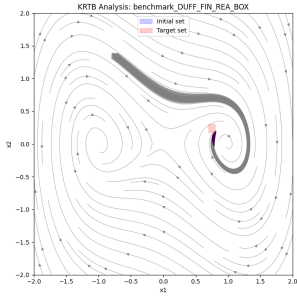
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -0.5x_2 - x_1(x_1^2 - 1) \end{bmatrix},$$

1.2 Benchmark DUFF_REA_BOX

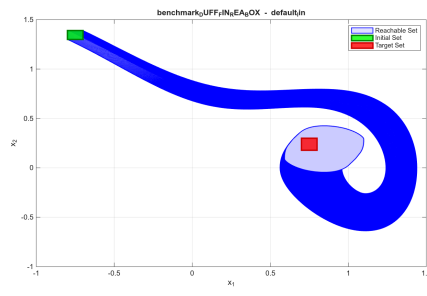
1.2.1 Specification

- Initial Set: $X_0 = [-0.8, -0.7] \times [1.3, 1.39]$
- Target Set: $X_F = [0.7, 0.8] \times [0.178, 0.3]$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

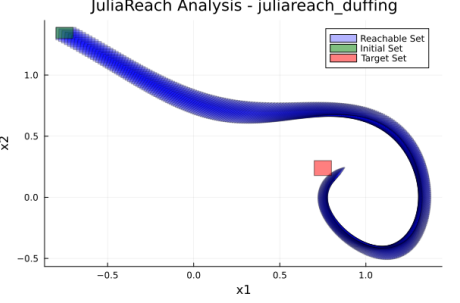
1.2.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark DUFF_REA_BOX.



(b) Result from Set-Propagation method (CORA) for benchmark DUFF_REA_BOX.



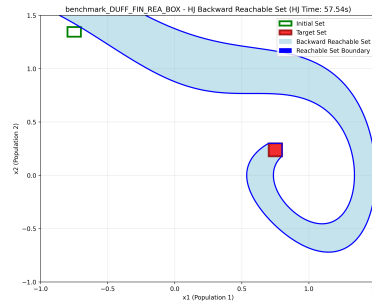
(c) Result from Set-Propagation method (JuliaReach) for benchmark DUFF_REA_BOX.

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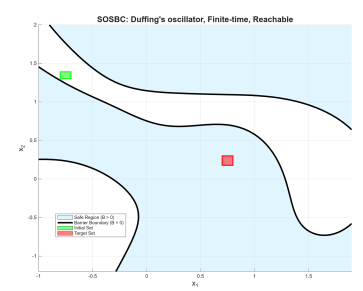
1 {
2   "timestamp": "2025-07-14T17:38:34.084357"
3   ,
4   "dreach.execution.time.seconds": 93.118,
5   "return_code": 0,
6   "stdout": "dReal Options:\nSMT:
7     duffing_fin_rea_box_0_0.smt2, PATH :
8     [1]\ndelta-sat with delta =
9     0.0010000000000000000\nFor k = 0, [1] --
10    SAT\n",
11   "stderr": "\nreal\tim33.118s\nuser\tim32
12   .990s\nsys\tim0.054s\n",
13   "success": true,
14   "tool": "dreach",
15   "setting": "setting_0",
16   "benchmark": "DUFF_FIN_REA_BOX",
17   "drh_file": "duffing_fin_rea_box.drh",
18   "container_runtime": "podman"
19 }

```

(d) Result from Counterexample Searching method (dreach) for benchmark DUFF_REA_BOX.



(e) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark DUFF_REA_BOX.



(f) Result from Barrier Certificate method (SOSTOOLS) for benchmark DUFF_REA_BOX (Infeasible Solution).

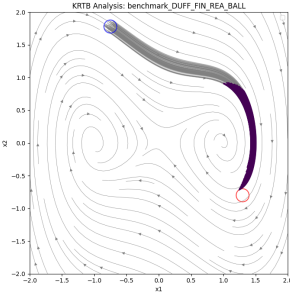
Figure 1: Results for the benchmark DUFF_REA_BOX.

1.3 Benchmark DUFF_REA_CVXLS

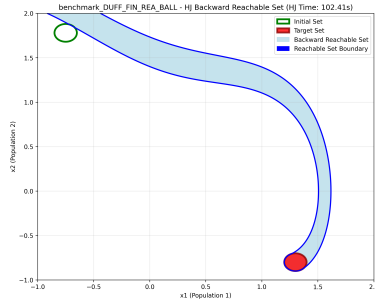
1.3.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^2 | (x_1 + 0.75)^2 + (x_2 - 1.78)^2 - 0.01 \leq 0\}$
- Target Set: $X_F = \{x \in \mathbb{R}^2 | (x_1 - 1.3)^2 + (x_2 + 0.8)^2 - 0.01 \leq 0\}$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

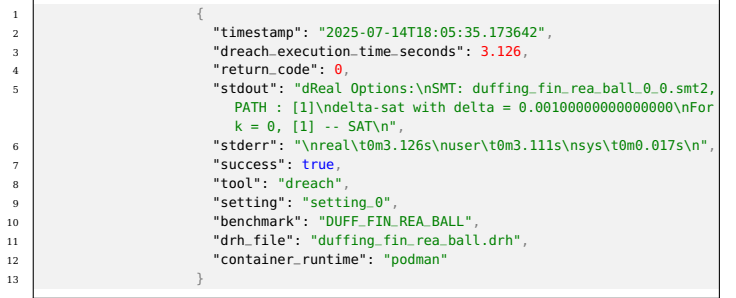
1.3.2 Results



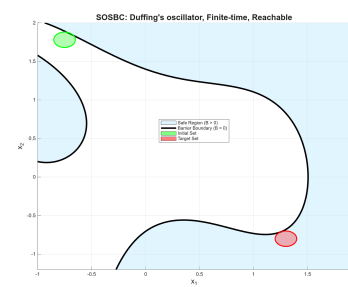
(a) Result from Reach-time Bounds method (Ours) for benchmark DUFF_REA_CVX_LS.



(c) Result from Hamilton-Jacobian PDE method (SOS-TOOLS) for benchmark DUFF_REA_CVXLS.



(b) Result from Counterexample Searching method (dreach) for benchmark DUFF_REA_CVX_LS.



(d) Result from Barrier Certificate method (SOSTOOLS) for benchmark DUFF_REA_CVXLS (**Infeasible Solution**).

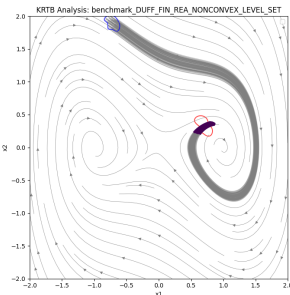
Figure 2: Results for the benchmark DUFF_REA_CVX_LS.

1.4 Benchmark DUFF_REA_NCVX_LS

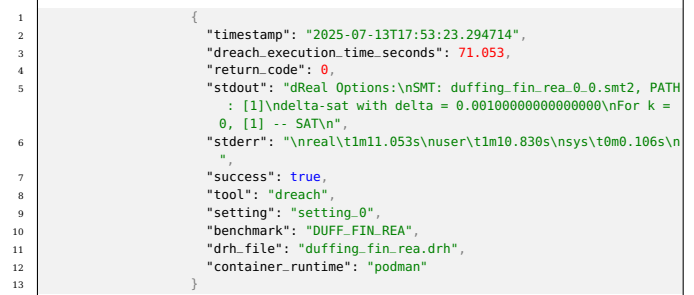
1.4.1 Specification

- **Initial Set:** $X_0 = \{x \in \mathbb{R}^2 | -\left(1 - \frac{x_1+0.75}{3} + \left(\frac{x_2-1.85}{0.05}\right)^5 + 2\left(\frac{x_1+0.75}{0.05}\right)^3\right) \exp\left(-\left[\left(\frac{x_1+0.75}{0.05}\right)^2 + \left(\frac{x_2-1.85}{0.05}\right)^2\right]\right) + 0.01 \leq 0\}$
- **Target Set:** $X_F = \{x \in \mathbb{R}^2 | -\left(1 - \frac{x_1-0.65}{3} + 3\left(\frac{x_2-0.25}{0.1}\right)^5 + 4\left(\frac{x_1-0.65}{0.1}\right)^3\right) \exp\left(-\left[\left(\frac{x_1-0.65}{0.1}\right)^2 + \left(\frac{x_2-0.25}{0.1}\right)^2\right]\right) + 0.9 \leq 0\}$
- **Time Horizon:** $[0, 6]$
- **Expected Result:** Reachable

1.4.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark DUFF REA NCVX LS.



(b) Result from Counterexample Searching method (dreach) for benchmark DUFF REA CVX LS.

Figure 3: Results for the benchmark DUFF REA NCVX LS.

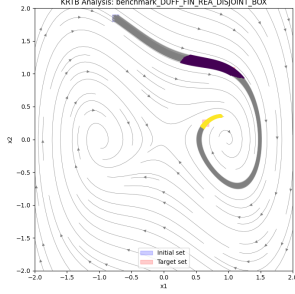
1.5 Benchmark DUFF_REA_DSJT_BOX

1.5.1 Specification

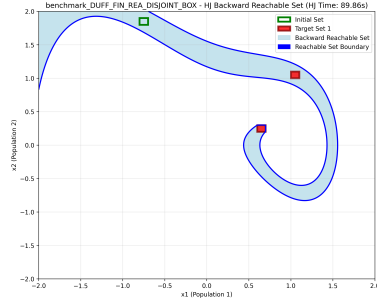
- Initial Set: $X_0 = [-0.8, -0.7] \times [1.8, 1.9]$

- Target Set: $X_F = [0.6, 0.7] \times [0.2, 0.3] \cup [1.0, 1.1] \times [1.0, 1.1]$
- Time Horizon: $[0, 6]$
- Expected Result: Reachable

1.5.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark DUFF_REA_DSJT_BOX.



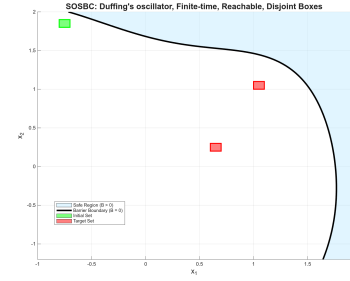
(c) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark DUFF_REA_DSJT_BOX.

```

1 {
2   "timestamp": "2025-07-15T14:14:45.368702",
3   "dreach_execution_time.seconds": 0.119,
4   "return_code": 0,
5   "stdout": "dReal Options:\nSMT: duffing_fin_rea_disjoint_box_0.0.smt2,\nPATH : [1,1]\nunsat\nFor k = 0, dReach tried 1 feasible paths, all of them are unsat.\nSMT: duffing_fin_rea_disjoint_box_1.0.smt2, PATH : [1,1]\nunsat\nFor k = 1, dReach tried 1 feasible paths, all of them are unsat.\nSMT: duffing_fin_rea_disjoint_box_2.0.smt2, PATH : [1,1]\nunsat\nFor k = 2, dReach tried 1 feasible paths, all of them are unsat.\nSMT: duffing_fin_rea_disjoint_box_3.0.smt2, PATH : [1,1]\nunsat\nFor k = 3, dReach tried 1 feasible paths, all of them are unsat.\n",
6   "stderr": "\nreal\t0m0.119s\nuser\t0m0.034s\nsys\t0m0.041s\n",
7   "success": true,
8   "tool": "dreach",
9   "setting": "setting_0",
10  "benchmark": "DUFF_FIN_REA_DISJOINT_BOX",
11  "drh_file": "duffing_fin_rea_disjoint_box.drh",
12  "container_runtime": "podman"
13 }

```

(b) Result from Counterexample Searching method (dreach) for benchmark DUFF_REA_CVX_LS.



(d) Result from Barrier Certificate method (SOSTOOLS) for benchmark DUFF_REA_DSJT_BOX (**Infeasible Solution**).

Figure 4: Results for the benchmark DUFF_REA_DSJT_BOX

1.6 Benchmark DUFF_UNR_BOX

1.6.1 Specification

- Initial Set: $X_0 = [-0.8, -0.7] \times [1.3, 1.35]$
- Target Set: $X_F = [0.7, 0.8] \times [0.178, 0.3]$
- Time Horizon: $[0, 20]$
- Expected Result: Unreachable

1.6.2 Results

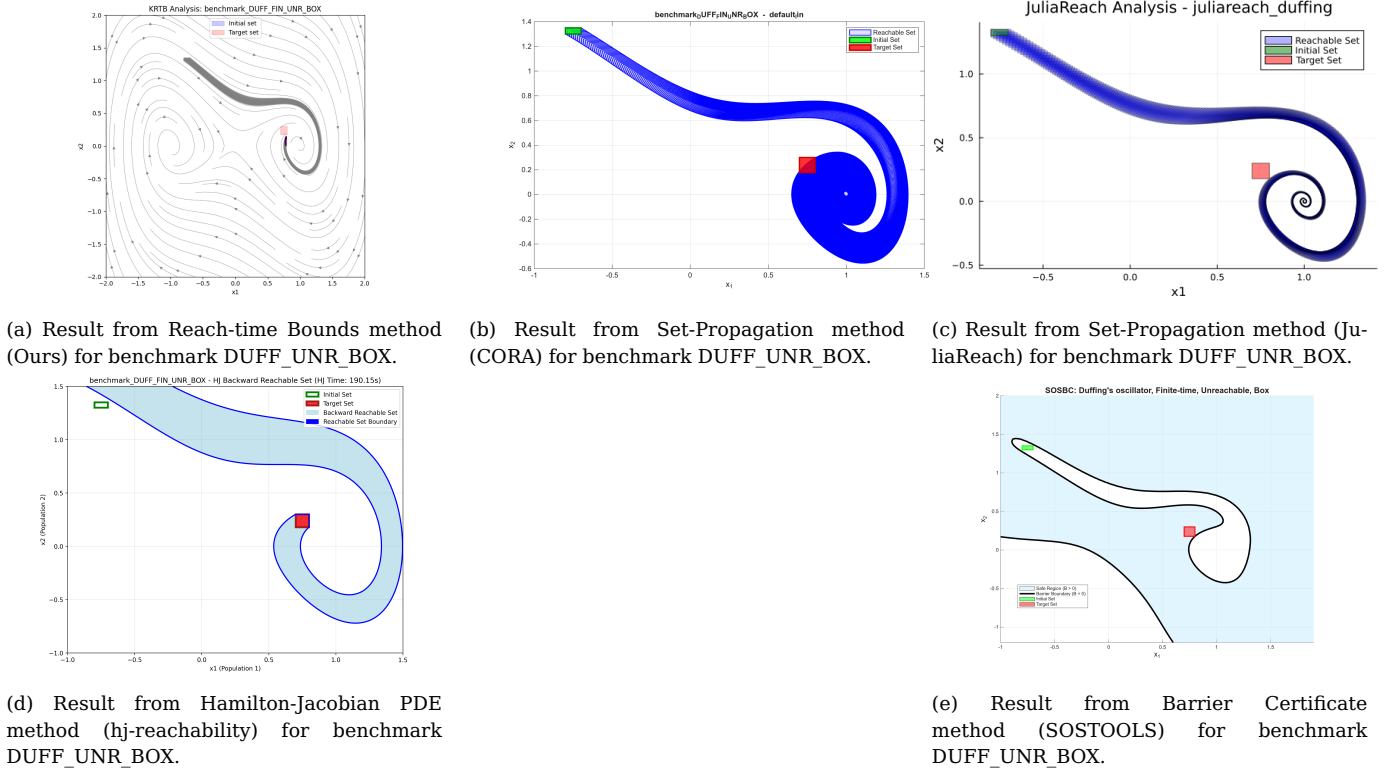


Figure 5: Results for the benchmark DUFF_UNR_BOX

1.7 Benchmark DUFF_UNR_CVX_LS

1.7.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^2 | (x_1 + 0.75)^2 + (x_2 - 1.75)^2 - 0.01 \leq 0\}$
- Target Set: $X_F = \{x \in \mathbb{R}^2 | (x_1 - 1.3)^2 + (x_2 + 0.8)^2 - 0.01 \leq 0\}$
- Time Horizon: $[0, 6]$
- Expected Result: Unreachable

1.7.2 Results

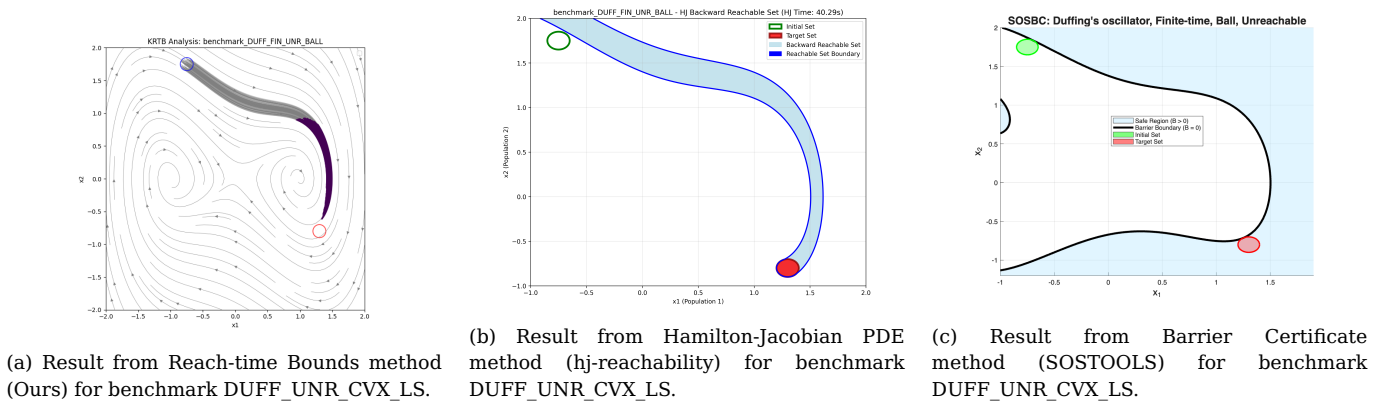


Figure 6: Results for the benchmark DUFF_UNR_CVX_LS

2 Nonlinear System with Stable Limit Cycle

2.1 System

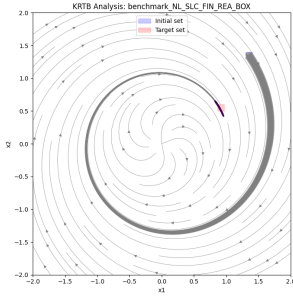
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} \frac{-x_1^3 + 4x_1^2x_2 - x_1x_2^2 + x_1 + 4x_2^3}{4(x_1^2 + x_2^2)} \\ \frac{-4x_1^3 - x_1^2x_2 - 4x_1x_2^2 - x_2^3 + x_2}{4(x_1^2 + x_2^2)} \end{bmatrix}$$

2.2 Benchmark NL-SLC_REA_BOX

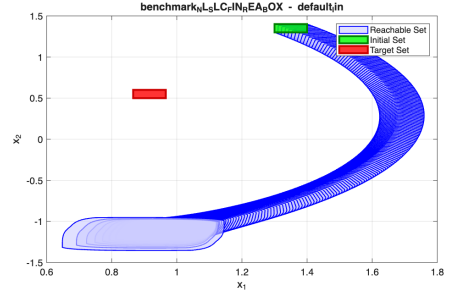
2.2.1 Specification

- Initial Set: $X_0 = [1.3, 1.4] \times [1.3, 1.4]$
- Target Set: $X_F = [\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} + 0.1] \times [0.5, 0.6]$
- Time Horizon: $[0, 15]$
- Expected Result: Reachable

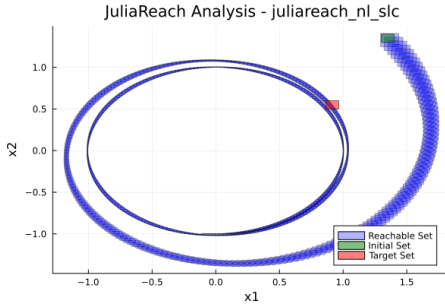
2.2.2 Results



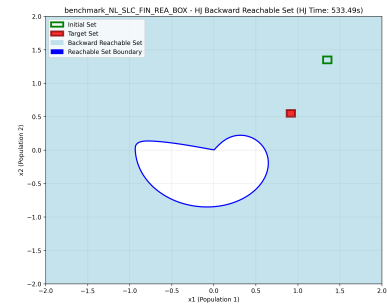
(a) Result from Reach-time Bounds method (Ours) for benchmark NL-SLC_REA_BOX.



(b) Result from Set-Propagation method (CORA) for benchmark NL-SLC_REA_BOX (Set Explosion).



(c) Result from Set-Propagation method (JuliaReach) for benchmark NL-SLC_REA_BOX.



(d) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark NL-SLC_REA_BOX.

Figure 7: Results for the benchmark NL-SLC_REA_BOX

Note on periodic reachability: The figure below further illustrates the reachable sets for Test Case 2.2.2 (a), computed using our method. Since the system has a stable limit cycle $\Gamma = \{x \mid \|x\| = 1\}$, trajectories that originated from the initial set X_0 may periodically pass through the target set X_F multiple times over some time horizon while converging to the limit cycle. To verify reachability under this challenging configuration, we consider the initial set $X_0 = [1.3, 1.4] \times [1.3, 1.4]$ and target set $X_F = [\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2} + 0.1] \times [0.5, 0.6]$. Applying our proposed method, we obtain the estimated reach-time bounds as $I_0 = [6.428, 6.625]$ and $I_1 = I_0 + P = [12.711, 12.908]$ where period $P = 6.283$. The visualization in Fig. 8 validates our estimations. We note that, as X_F approaches Γ , $\|\psi_1(X_F)\|$ and $\|\psi_2(X_F)\|$ approach zero, since the analytical principal eigenfunctions for this system are $\psi_1(x) = (1 - x_1^2 - x_2^2)^2 e^{\arctan(x_2/x_1)j}$ and $\psi_2(x) = \psi_1^*(x)$. Therefore, the reach time interval obtained using eigenfunction modulus approaches infinity. On the other hand, the reach time estimated using eigenfunction phase would be $I_{phase} = I_0 + mP$ with $m \in \mathbb{Z}$, which can intersect $I_{mag} = (-\infty, \infty)$ infinite number of times at periodic intervals, giving us periodic reach time estimates when X_F approaches the limit cycle Γ , as one would expect.

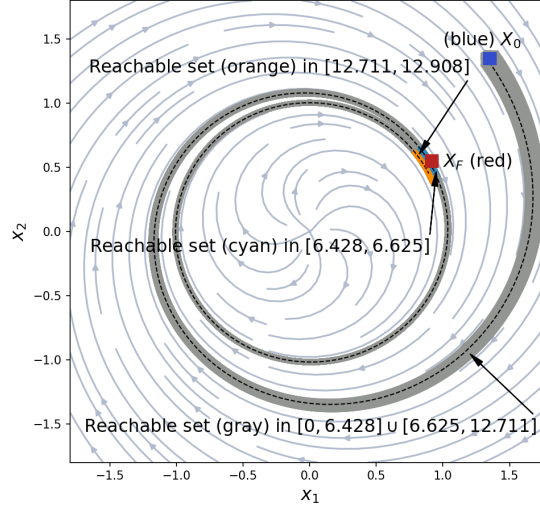
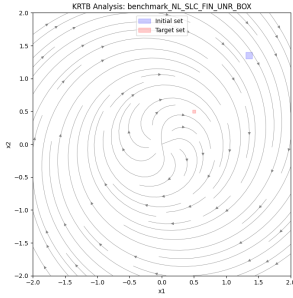


Figure 8: Reachable set of the nonlinear system with stable limit cycle from initial set X_0 for duration $[0, 6.428] \cup [6.625, 12.711]$ (gray), estimated reach-time bound $I_0 = [6.428, 6.625]$ (cyan), $I_1 = [12.711, 12.908]$ (orange) with a simulated periodic reachable trajectory (black dashed lines).

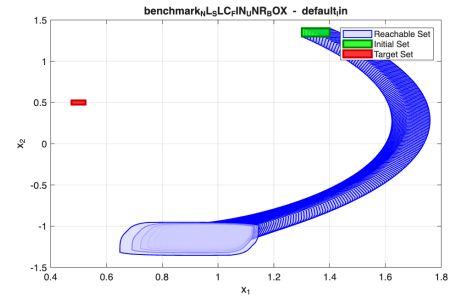
2.3 Benchmark NL-SLC_UNR_BOX

2.3.1 Specification

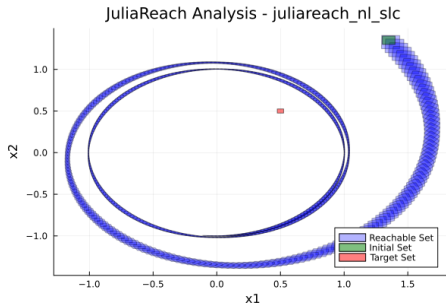
- Initial Set: $X_0 = [1.3, 1.4] \times [1.3, 1.4]$
- Target Set: $X_F = [0.475, 0.525] \times [0.475, 0.525]$
- Time Horizon: $[0, 15]$
- Expected Result: Unreachable



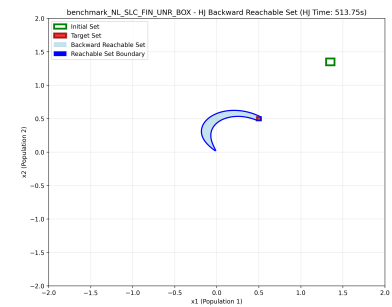
(a) Result from Reach-time Bounds method (Ours) for benchmark NL-SLC_UNR_BOX.



(b) Result from Set-Propagation method (CORA) for benchmark NL-SLC_REA_BOX (Set Explosion).



(c) Result from Set-Propagation method (JuliaReach) for benchmark NL-SLC_UNR_BOX.



(d) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark NL-SLC_REA_BOX.

Figure 9: Results for the benchmark NL-SLC_UNR_BOX

3 Nonlinear System with Known Eigenfunctions

3.1 System

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = [\nabla \Psi(x)]^{-1} \begin{bmatrix} -1 & 0 \\ 0 & 2.5 \end{bmatrix} \Psi(x)$$

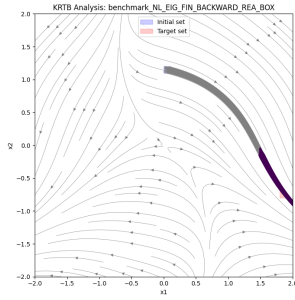
where the principal eigenfunctions are denoted as $\Psi(x) = [\psi_1(x), \psi_2(x)]^T$ with $\psi_1(x) = x_1^2 + 2x_2 + x_2^3$ and $\psi_2(x) = x_1 + \sin(x_2) + x_1^3$, and the associated eigenvalues are $\lambda_1 = -1$ and $\lambda_2 = 2.5$ at the unstable equilibrium $(0, 0)$.

3.2 Benchmark NL-EIG_BWD_REA_BOX

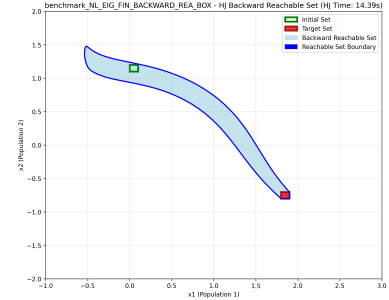
3.2.1 Specification

- Initial Set: $X_0 = [0, 0.1] \times [1.1, 1.2]$
- Target Set: $X_F = [1.8, 1.9] \times [-0.8, -0.7]$
- Time Horizon: $[0, 1.3]$
- Expected Result: Reachable

3.2.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark NL-EIG_BWD_REA_BOX.



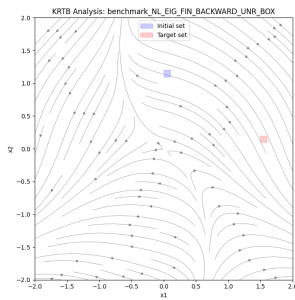
(b) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark NL-EIG_BWD_REA_BOX.

Figure 10: Results for the benchmark NL-EIG_BWD_UNR_BOX

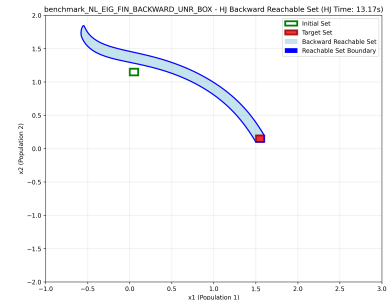
3.3 Benchmark NL-EIG_BWD_UNR_BOX

- Initial Set: $X_0 = [0, 0.1] \times [1.1, 1.2]$
- Target Set: $X_F = [1.5, 1.6] \times [0.1, 0.2]$
- Time Horizon: $[0, 1.3]$
- Expected Result: Reachable

3.3.1 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark NL-EIG_BWD_UNR_BOX.



(b) Result from Hamilton-Jacobian PDE method (hj-reachability) for benchmark NL-EIG_BWD_UNR_BOX.

Figure 11: Results for the benchmark NL-EIG_BWD_UNR_BOX

4 Cart-pole System

4.1 System

$$\dot{x} = \begin{bmatrix} \dot{p} \\ \dot{v} \\ \dot{\theta} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} v \\ \frac{[F + m_p \sin(\theta)(l\omega^2 + g \cos(\theta))]}{m_c + m_p \sin(\theta)^2} \\ \omega \\ \frac{[-F \cos(\theta) - m_p l \omega^2 \cos(\theta) \sin(\theta) - (m_c + m_p)g \sin(\theta)]}{l(m_c + m_p \sin(\theta)^2)} \end{bmatrix}$$

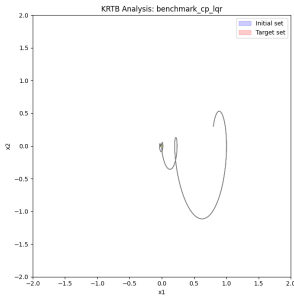
where $m_c = 2, m_p = 1, l = 1$, and $g = 9.8065$. The system equipped with the LQR controller $u(x) = [-3.162x_0 - 6.698x_1 - 5.027x_2 - 1.617x_3]$

4.2 Benchmark CP-LQR_REA_CVG

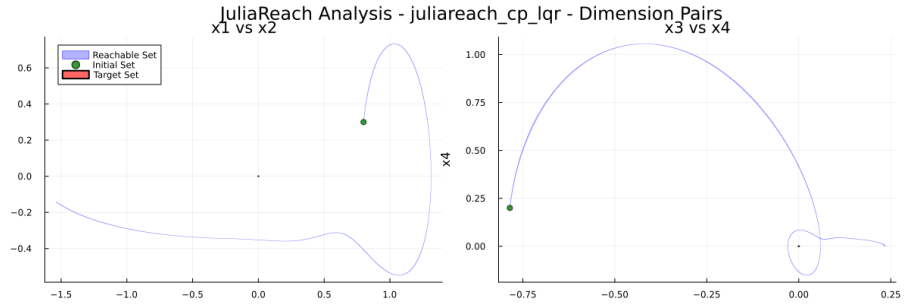
4.2.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid 10^{-4} \leq \|x\|_\infty \leq 10^{-2}, x \in X\}$
- Time Horizon: $[0, 10]$
- Expected Result: Reachable

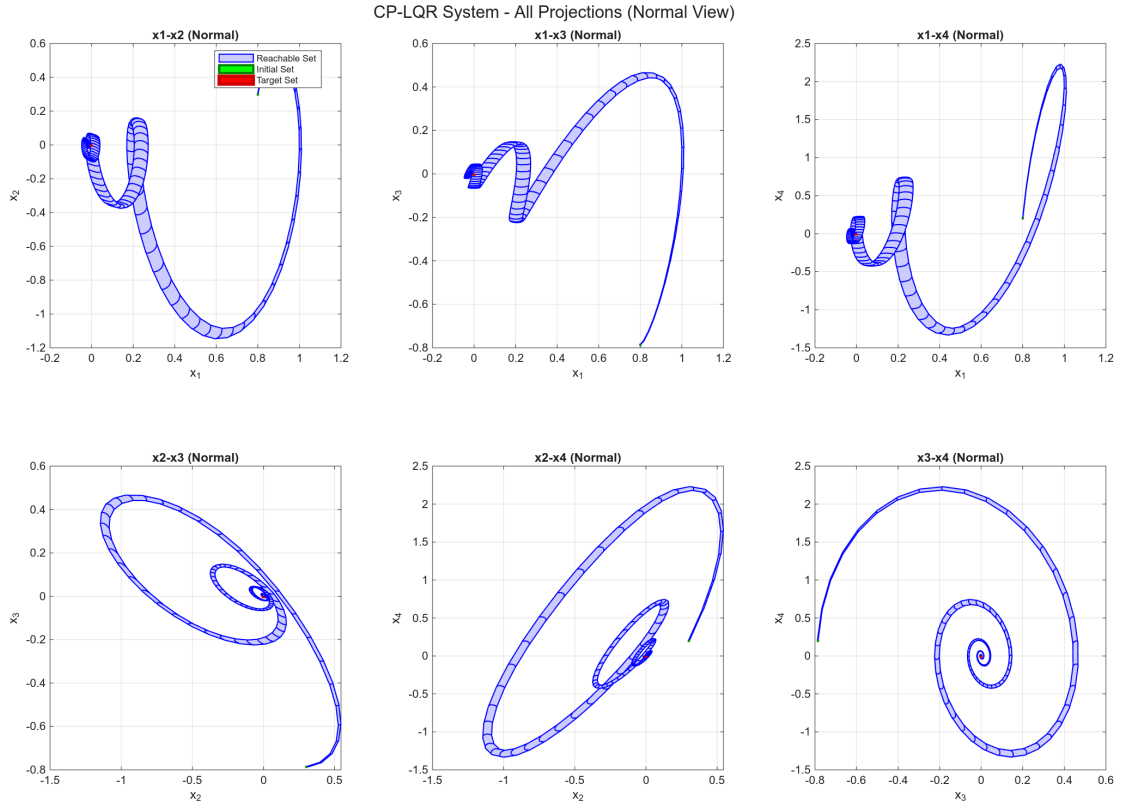
4.2.2 Results



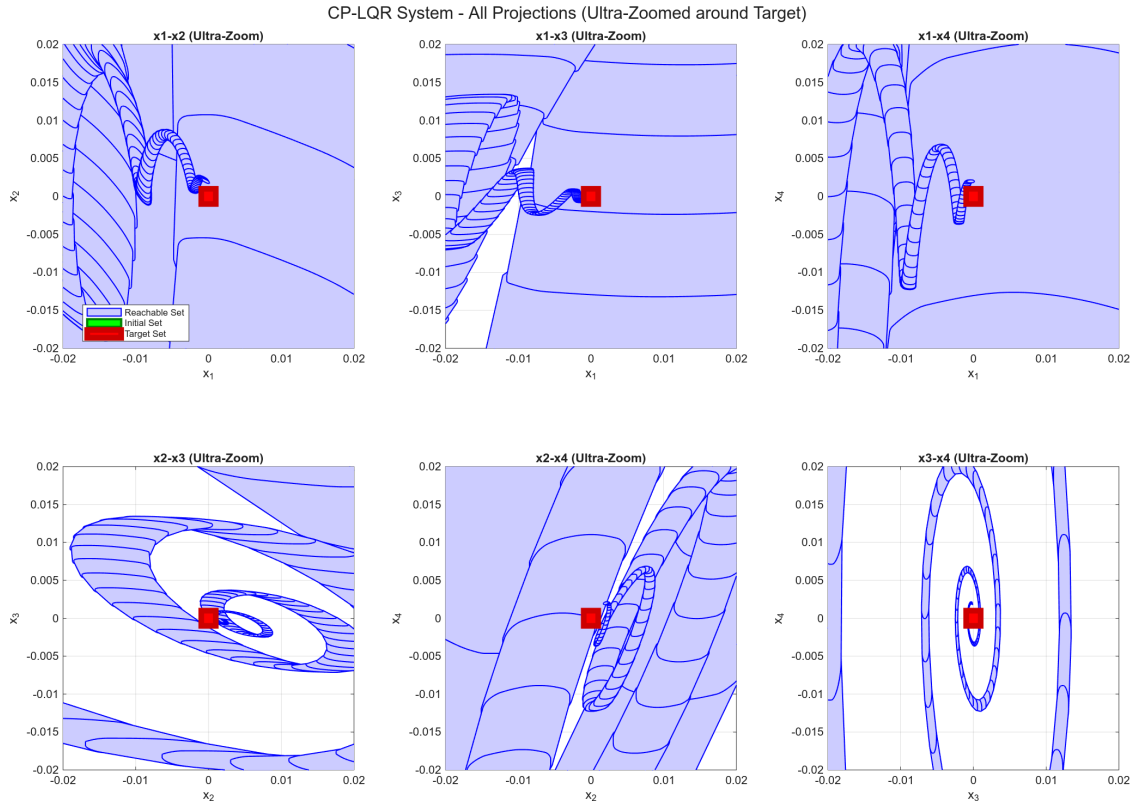
(a) Result from Reach-time Bounds method (Ours) for benchmark CP-LQR_REA_CVG.



(b) Result from Set-Propagation method (JuliaReach) for benchmark CP-LQR_REA_CVG. (Unsound Result)



(a) Result from Set-Propagation method (CORA) for benchmark CP-LQR_REA_CVG (normal scale).



(b) Result from Set-Propagation method (CORA) for benchmark CP-LQR_REA_CVG (zoomed scale). (Unsound Result)

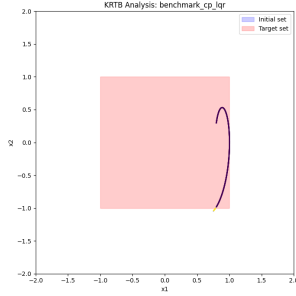
Figure 13: Results for the benchmark CP-LQR_REA_CVG

4.3 Benchmark CP-LQR_REA_USF

4.3.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid p + l \sin(\theta) > 1.3\}$
- Time Horizon: $[0, 10]$
- Expected Result: Reachable

4.3.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark CP-LQR_REA_USF.

```

1  {
2      "timestamp": "2025-07-14T12:46:56.358411",
3      "dreach_execution_time_seconds": 0.425,
4      "return_code": 0,
5      "stdout": "dReal Options:\nSMT: cp_lqr_fin_rea_unsafe_0_0.smt2, PATH :
6                  [1]\ndelta-sat with delta = 0.0010000000000000\nFor k = 0, [1] -- SAT\n",
7      "stderr": "\nreal\t0m0.425s\nuser\t0m0.348s\nsys\t0m0.019s\n",
8      "success": true,
9      "tool": "dreach",
10     "setting": "setting_0",
11     "benchmark": "CP_LQR_FIN_REA_UNSAFE",
12     "drh_file": "cp_lqr_fin_rea_unsafe.drh",
13     "container_runtime": "podman"
14 }

```

(b) Result from Counterexample Searching method (dreach) for benchmark CP-LQR_REA_USF.

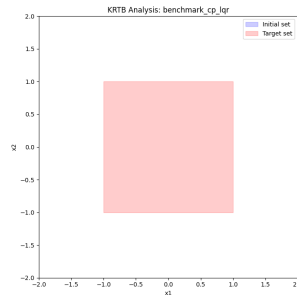
Figure 14: Results for the benchmark CP-LQR_REA_USF

4.4 Benchmark CP-LQR_UNR_SAF

4.4.1 Specification

- Initial Set: $X_0 = [0.8, 0.3, -\frac{\pi}{4}, 0.2]$
- Target Set: $X_F = \{x \mid p + l \sin(\theta) > 1.7\}$
- Time Horizon: $[0, 10]$
- Expected Result: Reachable

4.4.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark CP-LQR_REA_USF (No Feasible Trajectory).

Figure 15: Results for the benchmark CP-LQR_REA_USF

5 Multi-agent Consensus System

5.1 System

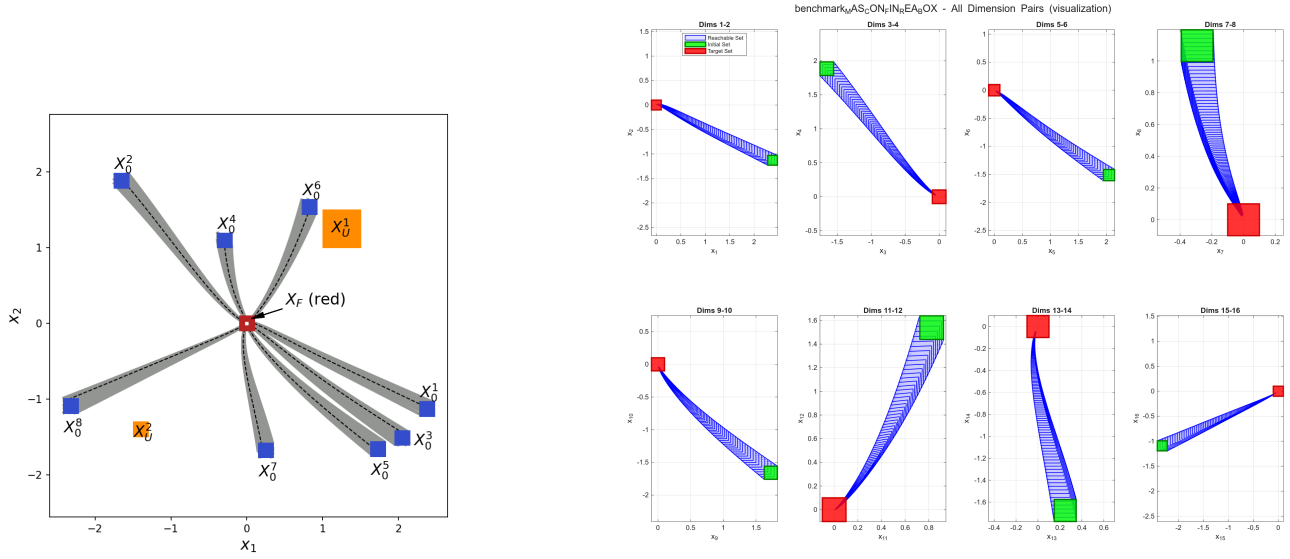
As defined in Fig. 16.

5.2 Benchmark MAS-CON_REA_BOX

5.2.1 Specification

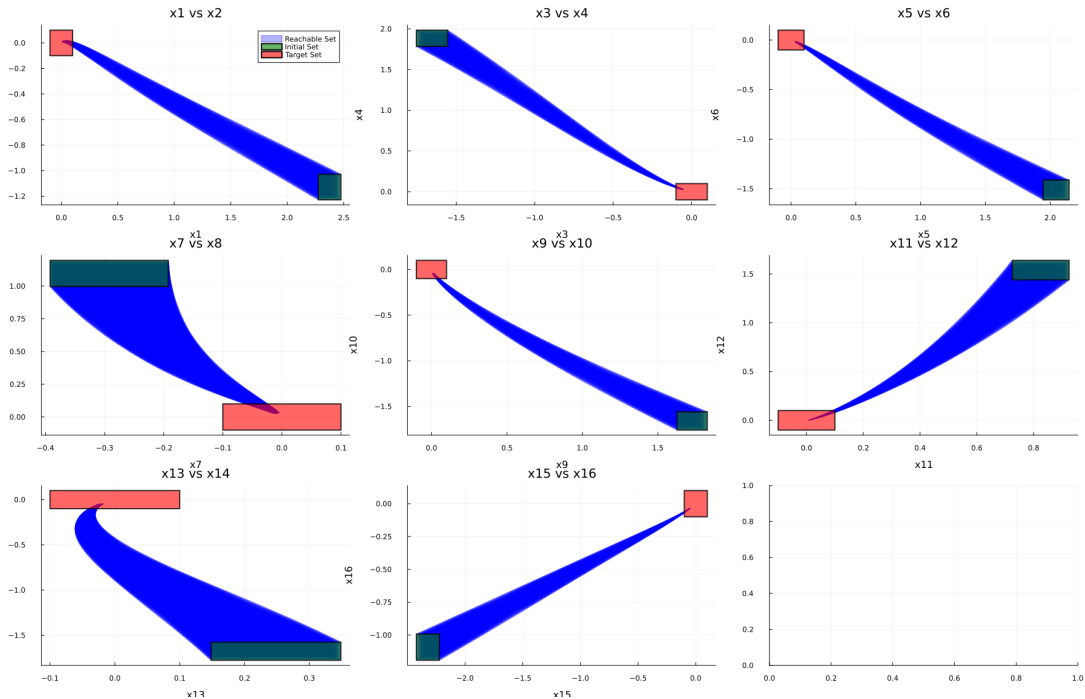
- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c\|_\infty \leq 0.1\}$,
where $c = [2.38, -1.13, -1.65, 1.89, 2.05, -1.51, -0.29, 1.10, 1.73, -1.66, 0.83, 1.54, 0.25, -1.68, -2.32, -1.09]^\top$
- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid 0.02 \leq \|x\|_\infty \leq 0.1, x \in X\}$
- Time Horizon: $[0, 7]$
- Expected Result: Reachable

5.2.2 Results



(a) Result from Reach-time Bounds method (Ours) for benchmark MAS-CON_REA_BOX.

(b) Result from Set-Propagation method (CORA) for benchmark MAS-CON_REA_BOX.
JuliaReach Analysis - juliareach_mas_con - Dimension Pairs



(c) Result from Set-Propagation method (JuliaReach) for benchmark MAS-CON_REA_BOX.

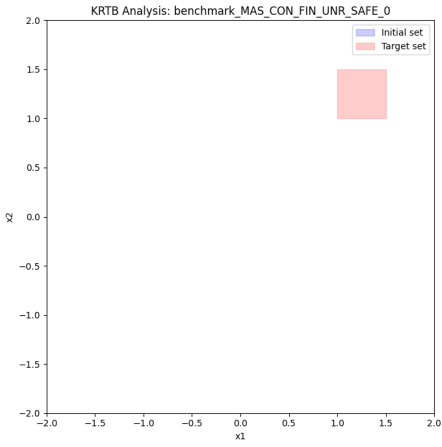
Figure 17: Results for the benchmark MAS-CON_REA_BOX

5.3 Benchmark MAS-CON_UNR_SAF_0

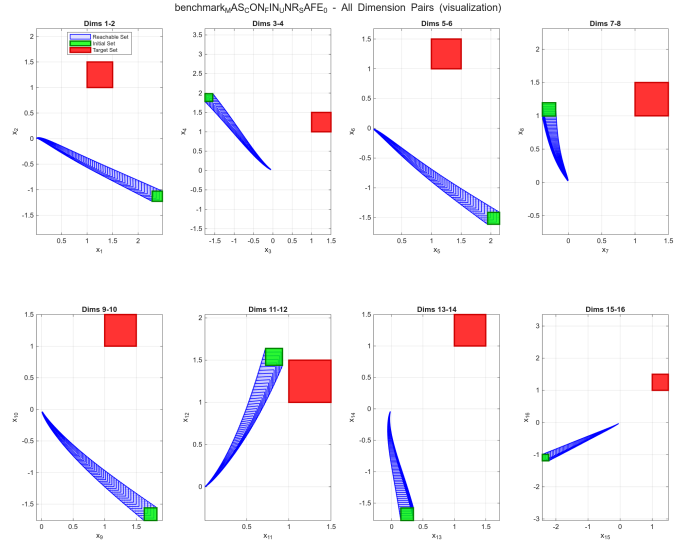
5.3.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c_0\|_\infty \leq 0.1\}$,
where $c_0 = [2.38, -1.13, -1.65, 1.89, 2.05, -1.51, -0.29, 1.10, 1.73, -1.66, 0.83, 1.54, 0.25, -1.68, -2.32, -1.09]^\top$
- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid \|x - c_F\|_\infty \leq 0.2, x \in X\}$,
where $c_F = [1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3]^\top$
- Time Horizon: $[0, 7]$
- Expected Result: Unreachable

5.3.2 Results

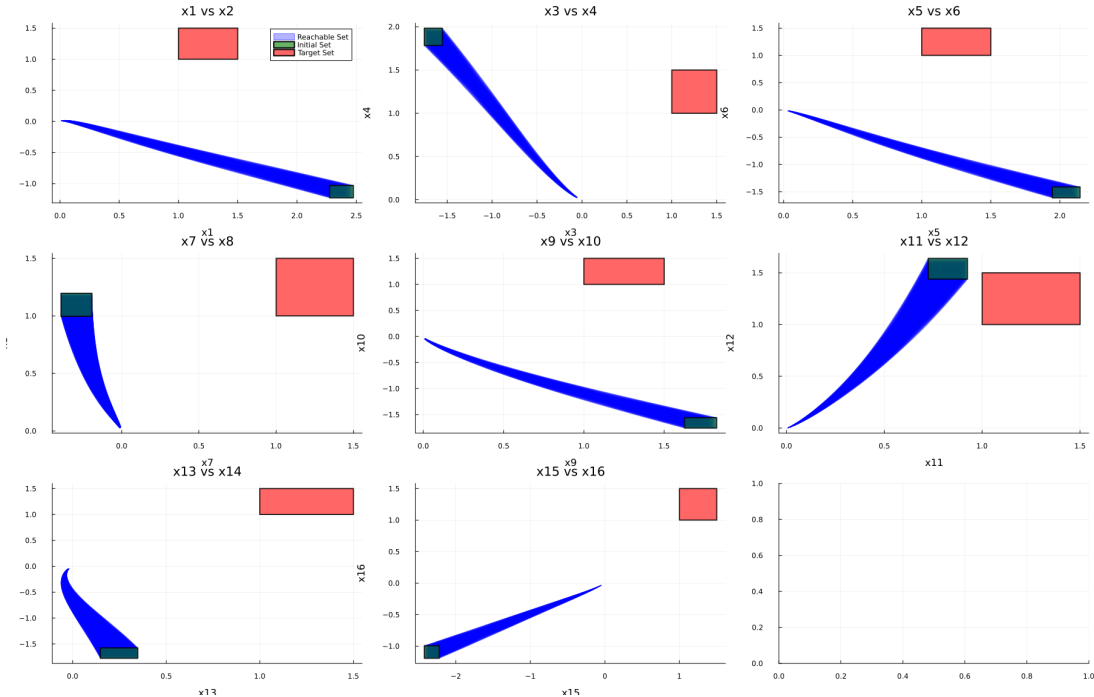


(a) Result from Reach-time Bounds method (Ours) for benchmark MAS-CON_UNR_SAF_0. (No Feasible Trajectory)



(b) Result from Set-Propagation method (CORA) for benchmark MAS-CON_UNR_SAF_0.

JuliaReach Analysis - juliareach_mas_con - Dimension Pairs



(c) Result from Set-Propagation method (JuliaReach) for benchmark MAS-CON_UNR_SAF_0.

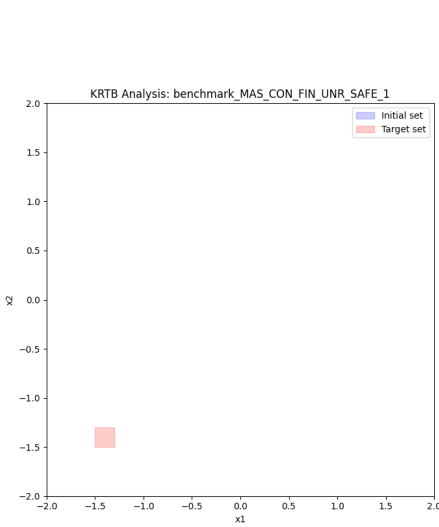
Figure 18: Results for the benchmark MAS-CON_UNR_SAF_0

5.4 Benchmark MAS-CON_UNR_SAF_1

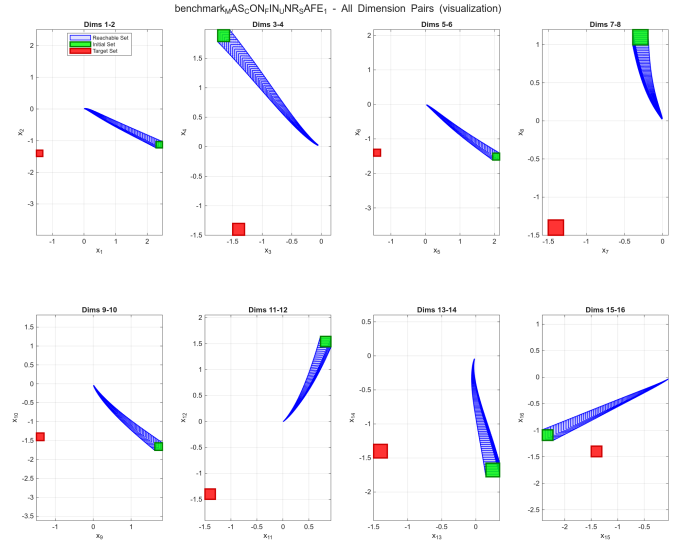
5.4.1 Specification

- Initial Set: $X_0 = \{x \in \mathbb{R}^{16} \mid \|x - c_0\|_\infty \leq 0.1\}$,
where $c_0 = [2.38, -1.13, -1.65, 1.89, 2.05, -1.51, -0.29, 1.10, 1.73, -1.66, 0.83, 1.54, 0.25, -1.68, -2.32, -1.09]^\top$
- Target Set: $X_F = \{x \in \mathbb{R}^{16} \mid \|x - c_F\|_\infty \leq 0.2, x \in X\}$,
where $c_F = [1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3, 1.3]^\top$
- Time Horizon: $[0, 7]$
- Expected Result: Unreachable

5.4.2 Results

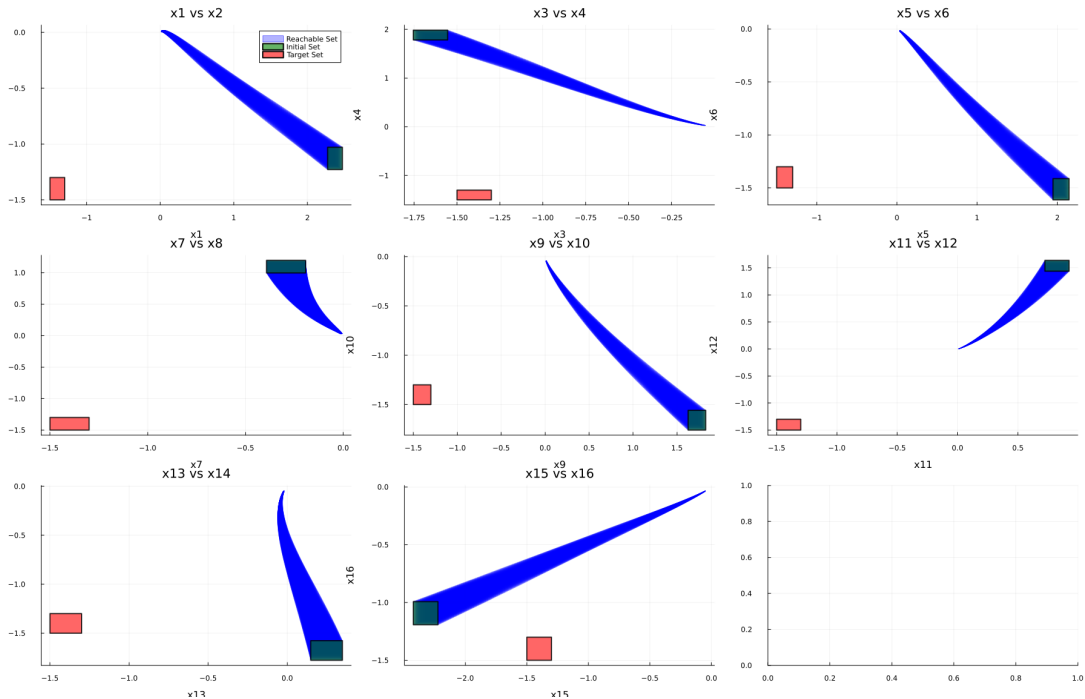


(a) Result from Reach-time Bounds method (Ours) for benchmark MAS-CON_UNR_SAF_1. (No Feasible Trajectory)



(b) Result from Set-Propagation method (CORA) for benchmark MAS-CON_UNR_SAF_1.

JuliaReach Analysis - juliareach_mas_con - Dimension Pairs



(c) Result from Set-Propagation method (JuliaReach) for benchmark MAS-CON_UNR_SAF_1.

Figure 19: Results for the benchmark MAS-CON_UNR_SAF_1

The dynamics of the i -th agent in the 16-agent system, for $i \in \{1, \dots, 16\}$, are governed by the general form:

$$\dot{x}_i = -\alpha x_i - \beta \cdot \frac{F_i(\mathbf{x})}{1 + e^{-x_i}} \quad (1)$$

where $\mathbf{x} = [x_1, \dots, x_{16}]^T$ is the joint state vector, $\alpha = 0.4419$ is the uniform self-damping rate, and $\beta = 0.1$ is the coupling strength. The nonlinear coupling functions $F_i(\mathbf{x})$, which define the network's topology and interaction laws, are explicitly defined below.

$F_1(\mathbf{x}) = 0.7799 \sin(x_1 - x_2) + 0.4384 \sin(x_1 - x_3) \\ + 0.7235 \sin(x_1 - x_4) + 0.5385 \sin(x_1 - x_6) \\ + 0.9359 \tanh(x_1 - x_2) + 0.5261 \tanh(x_1 - x_3) \\ + 0.8682 \tanh(x_1 - x_4) + 0.6462 \tanh(x_1 - x_6)$	$F_9(\mathbf{x}) = -0.9650 \sin(x_{11} - x_9) - 0.9450 \sin(x_{12} - x_9) \\ - 0.4723 \sin(x_{14} - x_9) + 0.2351 \sin(x_9 - x_{10}) \\ - 1.1580 \tanh(x_{11} - x_9) - 1.1341 \tanh(x_{12} - x_9) \\ - 0.5668 \tanh(x_{14} - x_9) + 0.2821 \tanh(x_9 - x_{10})$
$F_2(\mathbf{x}) = -0.2684 \sin(x_1 - x_2) + 0.8037 \sin(x_2 - x_4) \\ + 0.0659 \sin(x_2 - x_6) + 0.2881 \sin(x_2 - x_7) \\ + 0.9096 \sin(x_2 - x_8) - 0.3221 \tanh(x_1 - x_2) \\ + 0.9645 \tanh(x_2 - x_4) + 0.0791 \tanh(x_2 - x_6) \\ + 0.3458 \tanh(x_2 - x_7) + 1.0915 \tanh(x_2 - x_8)$	$F_{10}(\mathbf{x}) = -0.4858 \sin(x_{12} - x_{10}) - 0.3435 \sin(x_{14} - x_{10}) \\ - 0.3244 \sin(x_{15} - x_{10}) - 0.3004 \sin(x_{16} - x_{10}) \\ - 0.3087 \sin(x_9 - x_{10}) - 0.5830 \tanh(x_{12} - x_{10}) \\ - 0.4122 \tanh(x_{14} - x_{10}) - 0.3893 \tanh(x_{15} - x_{10}) \\ - 0.3605 \tanh(x_{16} - x_{10}) - 0.3705 \tanh(x_9 - x_{10})$
$F_3(\mathbf{x}) = -0.2134 \sin(x_1 - x_3) + 0.0249 \sin(x_3 - x_4) \\ + 0.2303 \sin(x_3 - x_7) - 0.2561 \tanh(x_1 - x_3) \\ + 0.0299 \tanh(x_3 - x_4) + 0.2764 \tanh(x_3 - x_7)$	$F_{11}(\mathbf{x}) = 0.7749 \sin(x_{11} - x_{12}) + 0.4606 \sin(x_{11} - x_{15}) \\ + 0.1655 \sin(x_{11} - x_9) + 0.9299 \tanh(x_{11} - x_{12}) \\ + 0.5528 \tanh(x_{11} - x_{15}) + 0.1986 \tanh(x_{11} - x_9)$
$F_4(\mathbf{x}) = -0.9091 \sin(x_1 - x_4) - 0.1332 \sin(x_2 - x_4) \\ - 0.5234 \sin(x_3 - x_4) + 0.6690 \sin(x_4 - x_5) \\ + 0.4908 \sin(x_4 - x_8) - 1.0910 \tanh(x_1 - x_4) \\ - 0.1598 \tanh(x_2 - x_4) - 0.6281 \tanh(x_3 - x_4) \\ + 0.8028 \tanh(x_4 - x_5) + 0.5889 \tanh(x_4 - x_8)$	$F_{12}(\mathbf{x}) = -0.8005 \sin(x_{11} - x_{12}) + 0.0407 \sin(x_{12} - x_{13}) \\ + 0.4758 \sin(x_{12} - x_{16}) + 0.8873 \sin(x_{12} - x_9) \\ + 0.6749 \sin(x_{12} - x_{10}) - 0.9606 \tanh(x_{11} - x_{12}) \\ + 0.0488 \tanh(x_{12} - x_{13}) + 0.5709 \tanh(x_{12} - x_{16}) \\ + 1.0647 \tanh(x_{12} - x_9) + 0.8099 \tanh(x_{12} - x_{10})$
$F_5(\mathbf{x}) = -0.8379 \sin(x_4 - x_5) + 0.3140 \sin(x_5 - x_6) \\ + 0.5726 \sin(x_5 - x_7) - 1.0055 \tanh(x_4 - x_5) \\ + 0.3768 \tanh(x_5 - x_6) + 0.6872 \tanh(x_5 - x_7)$	$F_{13}(\mathbf{x}) = -0.6587 \sin(x_{12} - x_{13}) + 0.3571 \sin(x_{13} - x_{14}) \\ + 0.8128 \sin(x_{13} - x_{15}) - 0.7905 \tanh(x_{12} - x_{13}) \\ + 0.4285 \tanh(x_{13} - x_{14}) + 0.9754 \tanh(x_{13} - x_{15})$
$F_6(\mathbf{x}) = -0.4528 \sin(x_1 - x_6) - 0.3530 \sin(x_2 - x_6) \\ - 0.4591 \sin(x_5 - x_6) + 0.4130 \sin(x_6 - x_7) \\ - 0.5434 \tanh(x_1 - x_6) - 0.4236 \tanh(x_2 - x_6) \\ - 0.5509 \tanh(x_5 - x_6) + 0.4956 \tanh(x_6 - x_7)$	$F_{14}(\mathbf{x}) = -0.0071 \sin(x_{13} - x_{14}) + 0.4631 \sin(x_{14} - x_{15}) \\ + 0.5999 \sin(x_{14} - x_9) + 0.7282 \sin(x_{14} - x_{10}) \\ - 0.0086 \tanh(x_{13} - x_{14}) + 0.5558 \tanh(x_{14} - x_{15}) \\ + 0.7198 \tanh(x_{14} - x_9) + 0.8738 \tanh(x_{14} - x_{10})$
$F_7(\mathbf{x}) = -0.7411 \sin(x_2 - x_7) - 0.4224 \sin(x_3 - x_7) \\ - 0.6344 \sin(x_5 - x_7) - 0.5229 \sin(x_6 - x_7) \\ + 0.0014 \sin(x_7 - x_8) - 0.8893 \tanh(x_2 - x_7) \\ - 0.5068 \tanh(x_3 - x_7) - 0.7613 \tanh(x_5 - x_7) \\ - 0.6275 \tanh(x_6 - x_7) + 0.0017 \tanh(x_7 - x_8)$	$F_{15}(\mathbf{x}) = -0.8285 \sin(x_{11} - x_{15}) - 0.1281 \sin(x_{13} - x_{15}) \\ - 0.2304 \sin(x_{14} - x_{15}) + 0.1325 \sin(x_{15} - x_{16}) \\ + 0.6078 \sin(x_{15} - x_{10}) - 0.9941 \tanh(x_{11} - x_{15}) \\ - 0.1538 \tanh(x_{13} - x_{15}) - 0.2765 \tanh(x_{14} - x_{15}) \\ + 0.1590 \tanh(x_{15} - x_{16}) + 0.7293 \tanh(x_{15} - x_{10})$
$F_8(\mathbf{x}) = -0.7094 \sin(x_2 - x_8) - 0.6962 \sin(x_4 - x_8) \\ - 0.0531 \sin(x_7 - x_8) - 0.8513 \tanh(x_2 - x_8) \\ - 0.8354 \tanh(x_4 - x_8) - 0.0638 \tanh(x_7 - x_8)$	$F_{16}(\mathbf{x}) = -0.7822 \sin(x_{12} - x_{16}) - 0.5326 \sin(x_{15} - x_{16}) \\ + 0.5749 \sin(x_{16} - x_{10}) - 0.9387 \tanh(x_{12} - x_{16}) \\ - 0.6392 \tanh(x_{15} - x_{16}) + 0.6900 \tanh(x_{16} - x_{10})$

Figure 16: The complete definition of the coupling functions $F_i(\mathbf{x})$ for the 16-agent system, as referenced in Eq. (1). The parameters are provided with full precision (rounded to four decimal places for display purposes).